

# CLARA DEHMAN

Department of Physics, University of Alicante, 03690 Alicante, Spain

🏠 Clara Dehman | 📞 (+34) 692 953 207 | ✉ clara.dehman@ua.es | 🆔 0000-0003-0554-7286

## Education

- Nov 2023: **Ph.D. in Physics**, Autonomous University of Barcelona, Spain — Cum Laude, International Mention
- Sep 2019: **Laurea Magistrale in Fisica**, University of Padua, Italy — Cum Laude (Erasmus Mundus NucPhys Program)
- Sep 2019: **European M.Sc. in Theoretical Nuclear Physics**, Spanish Consortium Universities, Spain — Matrícula de Honor (Erasmus Mundus NucPhys Program)
- Sep 2019: **M.Sc. in Physics**, University of Caen Normandy, France (Erasmus Mundus NucPhys Program)
- Jul 2017: **Bachelor's Degree in Physics**, Lebanese University, Lebanon — Ranked 1<sup>st</sup> in the Physics Department

## Research Positions

- Starting Nov 2026: **API Postdoctoral Researcher**, Anton Pannekoek Institute, Netherlands — Modeling Fast Radio Bursts
- Mar 2024–Oct 2026: **Juan de la Cierva Fellow**, University of Alicante, Spain — Chiral anomalies in magnetars
- Nov 2023–Feb 2024: **Postdoctoral Researcher**, Institute of Space Sciences, Spain — 3D MHD simulations of neutron stars
- Apr–Dec 2023 **Visiting Ph.D. Fellow**, Nordita, Sweden — Inverse cascade in neutron star crusts
- Nov 2019–Nov 2023: **Doctoral Researcher**, Institute of Space Sciences, Spain — Developed the 3D code *MATINS*
- Feb–Sep 2019: **Graduate Researcher**, University of Barcelona, Spain — Master's thesis on the BCPM equation of state
- Sep 2018–Jan 2019: **Intern**, Laboratoire de Physique Corpusculaire, France — Metamodeling for neutron star equations of state
- Feb–Jun 2018: **Intern**, Legnaro National Laboratories, Italy — Measured sub-barrier fusion cross sections

## Fellowships, Grants, & Funding

Total personal funding secured: €125,500 (excluding the declined €92,470 APOSTD Fellowship)

- Sept 2026–Aug 2029: **PID2025-171322NB-C21**, Agencia Estatal de Investigación, Spain — 200,000 €
- Feb 2025–Jan 2027: **Juan de la Cierva (JDC) Fellowship (PI)**, Agencia Estatal de Investigación, Spain — 72,000 €
- May 2026: **COST Action SCALES Travel Grant (PI)**, University of Coimbra, Portugal — 1,000 €
- Apr 2026: **International Visitor Grant (PI)**, Perimeter Institute, Canada — CAD 2,000
- Oct 2024–Oct 2026: **APOSTD Fellowship (PI)**, Generalitat Valenciana, Spain — 92,470 € (Declined in favor of JDC)
- Mar 2024–Dec 2026: **ASFAE/2022/26 Grant**, Generalitat Valenciana, Spain — 300,000 €
- Jan 2023–Dec 2026: **CIPROM 2022/13 Grant**, Generalitat Valenciana, Spain — 584,000 €
- Apr–Dec 2023: **Visiting Ph.D. Fellowship (PI)**, Nordita, Sweden — 80,000 SEK
- Apr 2022–Dec 2024: **SGR-Cat 2021 (01269) Grant**, Generalitat de Catalunya, Spain — 60,000 €
- Jan 2020–Feb 2024: **MAGNESIA ERC Grant**, European Research Council — 2,000,000 €
- Oct 2017–Sep 2019: **EACEA Erasmus Mundus Scholarship (PI)**, Joint Master Program — 37,000 €
- 2012–2014: **English Micro-Scholarship Program (PI)**, U.S. Embassy in Beirut — 2,000 \$

## Awards & Certificates

- Feb 2026: **Seal of Excellence**, European Commission — Score: 94.6/100
- Dec 2024: **IEEC Best Doctoral Thesis Award 2024** — Awarded 1,500 €
- May 2024: **Special Mention, SEA Thesis Award 2024** — Outstanding research in a highly competitive category
- Apr 2019: **Cambridge English: Advanced (CAE)**, Cambridge Assessment English (UK) — Grade C (183/210)
- Mar 2019: **Certificate in Data Analysis and Machine Learning**, Michigan State University & University of Oslo
- Apr 2017: **ERASMUS<sup>+</sup> Joint Master Program (NucPhys)** — Selected among the top 20% of international applicants

## Invited Talks & Seminars (20 total)

- Sep 2026: **Talk**, XII International Conference on Nuclear Physics in Astrophysics, Cluj-Napoca, [Romania](#)
- Jun 2026: **Review Talk**, SCALES Conference: Superfluid Condensates in Astrophysics and Laboratories, Coimbra, [Portugal](#)
- May 2026: **Seminar**, Astrophysical Fluid Dynamics Workshop, Escola d'Enginyeria de Barcelona, [Spain](#)
- Apr 2026: **Talk**, SCALES Conference: Vortex Motion from Laboratory to the Stars, Warsaw, [Poland](#) [[Link](#)]
- Apr 2026: **Strong Gravity Seminar**, Perimeter Institute, Waterloo, [Canada](#)
- Apr 2026: **Plasma-Astro Seminar**, Canadian Institute for Theoretical Astrophysics (CITA), Toronto, [Canada](#)
- Jun 2025: **Talk**, Fòrum IEEC 2025, Casa Convalescència (UAB), Barcelona, [Spain](#)
- May 2025: **Talk**, Extreme Physics of Neutron Star Interiors, Princeton University, [USA](#)
- May 2025: **Review Talk**, Workshop on Modern Equation of State and Spectroscopy in Neutron Stars, Alcalá University, [Spain](#)
- Feb 2025: **Talk**, Breaking New Ground in Supernova Physics, Fukuoka University, [Japan](#)
- Dec 2024: **Review Talk**, IReNA-INT Joint Workshop on Thermal and Magnetic Evolution, University of Washington, [USA](#)
- Sep 2024: **Talk**, Frontier Workshop in Astrophysics IV, Palermo, [Italy](#)
- Jun 2023: **Astrophysics Seminar**, University of Valencia, [Spain](#)
- May 2023: **Astrophysics Seminar**, Nordita, [Sweden](#) [[Link](#)]
- Jun 2022: **Astrocoffee Seminar**, Goethe University, [Germany](#) [[Link](#)]
- May 2022: **Seminar**, NucPhys Master Program, GANIL, [France](#)

- Dec 2021: **Seminar**, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, [Spain](#) [[Link](#)]
- Nov 2021: **Astrophysics Seminar**, University of Alicante, [Spain](#)
- Dec 2020: **Talk**, Stars & Compact Objects Meeting, Flatiron Institute, New York, [USA](#)
- Mar 2019: **Seminar**, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, [Spain](#)

## Invited Scientific Visit

- Apr 2026: **Invited Scientific Visit**, Perimeter Institute for Theoretical Physics, Ontario, [Canada](#)

## Contributed Talks & Seminars (13 total)

- Jun 2026: **Talk**, The X-ray Universe 2026, Elche, [Spain](#)
- Jun 2026: **Talk**, NewAthena Rising, ICE-CSIC Barcelona, [Spain](#)
- Jun 2025: **Talk**, EAS Annual Meeting, University College Cork, [Ireland](#)
- Jul 2024: **Talk**, 10<sup>th</sup> International Conference on Quarks and Nuclear Physics, University of Barcelona, [Spain](#)
- Jun 2024: **Talk**, XMM-Newton Workshop 2024, European Space Agency, Madrid, [Spain](#)
- May 2024: **Special Seminar**, Institute of Space Sciences, Barcelona, [Spain](#)
- Apr 2024: **Astrophysics Seminar**, University of Alicante, [Spain](#)
- May 2022: **Talk**, Pharos Conference, University of La Sapienza, Rome, [Italy](#)
- Mar 2022: **Seminar**, Institute of Space Sciences, Barcelona, [Spain](#) [[Link](#)]
- Nov 2021: **Talk**, IAU Symposium 363 (online)
- Jun 2021: **Talk**, EAS Annual Meeting (online)
- Nov 2020: **Seminar**, Institute of Space Sciences, Barcelona, [Spain](#) [[Link](#)]
- Mar 2020: **Talk**, PHAROS Conference, Patras, [Greece](#) (canceled due to COVID-19)

## Software Development & Open Access Tools

Since Sep 2020: Lead developer of **MATINS** — a 3D code for the MAGneto-**T**hermal evolution in Isolated Neutron Stars — [MATINS webpage](#)

- *Open access:* [github.com/ice-csic-astroexotic/MATINS](https://github.com/ice-csic-astroexotic/MATINS)
- *Lead developers:* [C. Dehman](#), D. Viganò, S. Ascenzi
- *Implementation:* Written in Fortran 90 with OpenMP parallelization, optimized for Intel compilers (better performance than GNU).



## Mentoring

### Supervision

- Since Sept 2026: **Supervisor**, B.Sc. student, University of Alicante, [Spain](#) — Ahlam Makboul
- Since Mar 2025: **Co-supervisor**, Ph.D. student, University of Alicante, [Spain](#) — Joan Llorens Ripoll
- Oct 2025–Apr 2026: **Research mentor**, M.Sc. student, Institute of Space Sciences, [Spain](#) — Riccardo Ricci
- Jan–Jul 2021: **Research mentor**, M.Sc. student, Institute of Space Sciences, [Spain](#) — Marco Ermolli

### Teaching (112 Hours Total)

- 2024–2025: **Instructor**, Introduction to Modeling in Physics, University of Alicante, [Spain](#) — 45 hr (95 B.Sc. students)
- 2024–2025: **Instructor**, Nuclear and Particle Physics, University of Alicante, [Spain](#) — 54 hr (112 B.Sc. students)
- Jan–Mar 2025: **Lecturer**, Stellar Astrophysics, University of Alicante, [Spain](#) — 12 hr (14 B.Sc. students)
- May 2025: **Invited Lecturer**, Research in Computational Astrophysics, University of Alicante, [Spain](#) — 1 hr (10 M.Sc. students)

## Scientific Community Service

### Scientific Meeting Organization

- Sep 2026: **SOC & LOC**, 22<sup>nd</sup> Pencil Code User Meeting, University of Alicante, [Spain](#) [[Link](#)]
- Nov 2022: **LOC**, Athena X-ray Observatory Conference, Barcelona, [Spain](#) [[Link](#)]
- May 2022: **LOC**, PHAROS Final Conference on Neutron Stars, University of La Sapienza, Rome, [Italy](#) [[Link](#)]

### Reviewing & Evaluation Activities

- Jan 2026: **Ph.D. Committee Member**, Juan A. Gil Granados, University of Barcelona, [Spain](#) [[Link](#)]
- Since Jun 2025: **Expert Evaluator**, Ph.D. proposals, Valencian Agency for Assessment and Prospective, [Spain](#) (2 proposals)
- Since 2022: **Journal Reviewer**, Nat. Astron.; Nat. Commun.; ApJ; MNRAS; J. Phys. G

## Outreach

- Jun 2026: **Social Media Outreach Coordinator**, SCALES COST Action CA24139 (WP4)
- Feb 2026: **Invited Outreach Talk**, International Day of Girls and Women in Science, University of Alicante, [Spain](#)
- Jun 2024: **Invited Interview**, on how scientific discoveries will shape the future of science, University of Elche, [Spain](#) [[Link](#)]
- Dec 2023: **Invited Interview**, on my Ph.D. experience, ICE-CSIC, Barcelona, [Spain](#) [[Interview](#) | [Newsletter](#)]
- Nov 2022: **Astronomical Observation Event**, public stargazing event, Autonomous University of Barcelona, [Spain](#) [[Link](#)]
- Sep 2022: **Research Night**, astronomical observation event for school children, Igualada, [Spain](#)
- Feb 2022: **Outreach Talk**, International Day of Girls and Women in Science, ICE-CSIC, [Spain](#) [[Link](#)]
- 2021–2023: **Magnet ICE Outreach**, collaboration with teachers at Gabriel Castellà i Raich to develop science projects promoting student integration in a school facing segregation challenges, Igualada, [Spain](#)

## Press Releases (14 Total)

---

- Jul 2024: **Three unusually cold young neutron stars discovered**, Europa Press [\[Link\]](#)
- Jul 2024: **Unusually cold neutron stars detected by UA researchers**, TodoAlicante [\[Link\]](#)
- Jul 2024: **UA researchers contribute to discovery of unusually cold neutron stars**, University of Alicante [\[Link\]](#)
- Jun 2024: **Chandra peers into the densest and weirdest stars**, NASA [\[Link\]](#)
- Jun 2024: **Too young to be so cool: lessons from three neutron stars**, ESA [\[Link\]](#)
- Jun 2024: **Three neutron stars too cold for their age discovered**, IEEC [\[Link\]](#)
- Jun 2024: **Too young to be so cold**, MEDIA INAF [\[Link\]](#)
- Jun 2024: **Three neutron stars too cold for their age discovered**, ICE-CSIC [\[Link\]](#)
- Dec 2022: **Outburst of the young magnetar Swift J1818.0–1607 observed**, Phys.org [\[Link\]](#)
- Jun 2020: **Youngest neutron star discovered**, The Independent [\[Link\]](#)
- Jun 2020: **Youngest “baby” dead star discovered**, CNET [\[Link\]](#)
- Jun 2020: **Baby pulsar and rare magnetar discovered**, Astronomy Now [\[Link\]](#)
- Jun 2020: **Youngest magnetar ever discovered**, Sci.News [\[Link\]](#)
- Jun 2020: **Discovery of a “baby” star explaining cosmic explosions**, La Vanguardia [\[Link\]](#)

## Collaborations & Memberships

---

- Since Jul 2026: **Junior Member**, International Astronomical Union (IAU)
- Since Nov 2025: **Member**, SCALES European COST Action CA24139
- Since Mar 2025: **Member**, NewAthena Science Community
- Since Sep 2022: **Member**, Einstein Telescope Collaboration
- Since Mar 2022: **Member**, AIHUB — Artificial Intelligence Initiative of the Spanish National Research Council (CSIC)
- 2019–2022: **Member**, Pharos European COST Action CA16214

## Languages

---

- **Arabic**: Native
- **English**: Fluent (C2; Cambridge Advanced 183/210, TOEFL 805/900)
- **French**: Fluent (C2)
- **Spanish**: Intermediate (B2)
- **Italian**: Intermediate (B2)

## List of Publications

---

Curated selection of 23 peer-reviewed publications: 8 as first author, 2 as co-first author (including a *Nature Astronomy* article), and one invited Living Review. Full publication list available via [ADS](#), [arXiv](#), [ORCID](#), and [Google Scholar](#). **Citations**: 660+ | **h-index**: 15

**23.** M. Ronchi, C. Pardo Araujo, V. Graber, N. Rea, [C. Dehman](#) et al. — *Connecting radio pulsars, magnetars and XDINSs in a unified evolutionary framework, using simulation-based inference*. Under minor revision in *ApJ*, 2026 [\[PDF\]](#)

**22.** [C. Dehman](#) — *Magnetar field dynamics driven by chiral anomalies without magnetic helicity*. In press, *Phys. Rev. D*, 2026 [\[arXiv:2605.08068\]](#) | [DOI](#)

**21.** J.A. Pons, [C. Dehman](#), D. Viganò: 2026. *Magnetic, thermal and rotational evolution of isolated neutron stars*. *Living Rev Comput Astrophys* 12, 5. [\[arXiv:2509.06699\]](#) | [DOI](#) | cit: 142

**20.** A. Suvorov, [C. Dehman](#)\*, J.A. Pons: 2026. *Late-blooming magnetars: awakening as ultra-long period objects after a dormant cooling epoch*. *ApJ* 1000, 55 (equal contribution by first two authors; CA: [C. Dehman](#)) [\[arXiv:2505.05373\]](#) | [DOI](#) | cit: 7

**19.** A. Suvorov, [C. Dehman](#), J.A. Pons: 2025. *Revealing the nature of ultra-long period objects with space-based gravitational-wave interferometers*. *ApJ* 991, 134 [\[arXiv:2505.06125\]](#) | [DOI](#) | cit: 6

**18.** [C. Dehman](#) and J.A. Pons: 2025. *Magnetar field dynamics shaped by chiral anomalies and helicity*. *Phys. Rev. Research* 7, 033231 [\[arXiv:2505.06196\]](#) | [DOI](#) | cit: 8

**17.** [C. Dehman](#) and A. Brandenburg: 2025. *Reality of inverse cascading in neutron star crusts*. *A&A* 694, A39 [\[arXiv:2408.08819\]](#) | [DOI](#) | cit: 9

**16.** S. Ascenzi, D. Viganò, [C. Dehman](#) et al.: 2024. *3D code for Magneto–Thermal evolution in Isolated Neutron Stars, MATINS: thermal evolution and light curves*. *MNRAS* 533, 201 [\[arXiv:2401.15711\]](#) | [DOI](#) | cit: 27

**15.** [C. Dehman](#), M. Centelles, X. Viñas: 2024. *Impact of the hot inner crust on compact stars at finite temperature*. *A&A* 687, A236 [\[arXiv:2401.16957\]](#) | [DOI](#) | cit: 7

**14.** A. Marino\*, [C. Dehman](#)\*, K. Kovlakas\*, N. Rea\* et al.: 2024. *Constraints on the dense matter equation of state from young and cold isolated neutron stars*. *Nature Astron.* 8, 1020 (equal contribution by these authors) [\[arXiv:2404.05371\]](#) | [DOI](#) | cit: 32

**13.** J. Urbà, P. Stefanou, [C. Dehman](#), J.A. Pons: 2023. *Modeling force-free neutron star magnetospheres using physics-informed neural networks*. *MNRAS* 524, 32 [\[arXiv:2303.11968\]](#) | [DOI](#) | cit: 20

**12.** [C. Dehman](#), D. Viganò, S. Ascenzi et al.: 2023. *3D evolution of neutron star magnetic-fields from a realistic core-collapse turbulent topology*. *MNRAS* 523, 5198 [\[arXiv:2305.06342\]](#) | [DOI](#) | cit: 31

**11.** [C. Dehman](#), J.A. Pons, D. Viganò, N. Rea: 2023. *How bright can old magnetars be? Assessing the impact of magnetized envelopes and field topology on neutron star cooling*. *MNRAS Lett.* 520, L42 [\[arXiv:2301.02261\]](#) | [DOI](#) | cit: 15

10. A.Y. Ibrahim, A. Borghese, N. Rea et al.: 2023. *Deep X-ray and radio observations of the first outburst of the young magnetar Swift J1818.0-1607*. ApJ 943,20 [[arXiv:2211.12391](#) | [DOI](#)] cit: 16]
9. C. Dehman, D. Viganò, J.A. Pons, N. Rea: 2023. *3D code for MAGneto-thermal evolution in Isolated Neutron Stars, MATINS: the magnetic field formalism*. MNRAS 518, 1222 [[arXiv:2209.12920](#)] [DOI](#) | cit: 39]
8. N. Rea, F. Coti Zelati, C. Dehman et al.: 2022. *Constraining the nature of the 18 min periodic radio transient GLEAM-X J162759.5-523504.3 via multi-wavelength observations and magneto-thermal simulations*. ApJ 940, 72 [[arXiv:2210.01903](#)] [DOI](#) | cit: 40]
7. F. Anzuini, A. Melatos, C. Dehman et al.: 2022. *Thermal luminosity degeneracy of magnetized neutron stars with and without hyperon cores*. MNRAS 515, 3014 [[arXiv:2205.14793](#)] [DOI](#) | cit: 11]
6. F. Anzuini, A. Melatos, C. Dehman et al.: 2022. *Fast cooling and internal heating in hyperon stars*. MNRAS 509, 2609 [[arXiv:2110.14039](#)] [DOI](#) | cit: 28]
5. D. Viganò, A. Garcia-Garcia, J.A. Pons, C. Dehman et al.: 2021. *Magneto-thermal evolution of neutron stars with coupled Ohmic, Hall and ambipolar effects via accurate finite-volume simulations*. Comput. Phys. Commun. 265, 108001 [[arXiv:2104.08001](#)] [DOI](#) | cit: 61]
4. A.M. Stefanini, G. Montagnoli, M. D'Andrea, M. Giacomini, C. Dehman et al.: 2021. *New insights into sub-barrier fusion of  $^{28}\text{Si} + ^{100}\text{Mo}$* . J. of Phys. G 48, 055101 [[ADS](#)] [DOI](#) | cit: 21]
3. F. Coti Zelati, A. Borghese, G.L. Israel et al.: 2021. *The new magnetar SGR J1830-0645 in outburst*. ApJ Lett. 907, L34 [[arXiv:2011.08653](#)] [DOI](#) | cit: 24]
2. C. Dehman, D. Viganò, N. Rea et al.: 2020. *On the rate of crustal failures in young magnetars*. ApJ Lett. 902, L32 [[arXiv:2010.00617](#)] [DOI](#) | cit: 43]
1. P. Esposito, N. Rea, A. Borghese et al.: 2020. *A very young radio-loud magnetar*. ApJ Lett. 896, L30 [[arXiv:2004.04083](#)] [DOI](#) | cit: 80]